Turbulence and its nonlinear cross-correlation analyses

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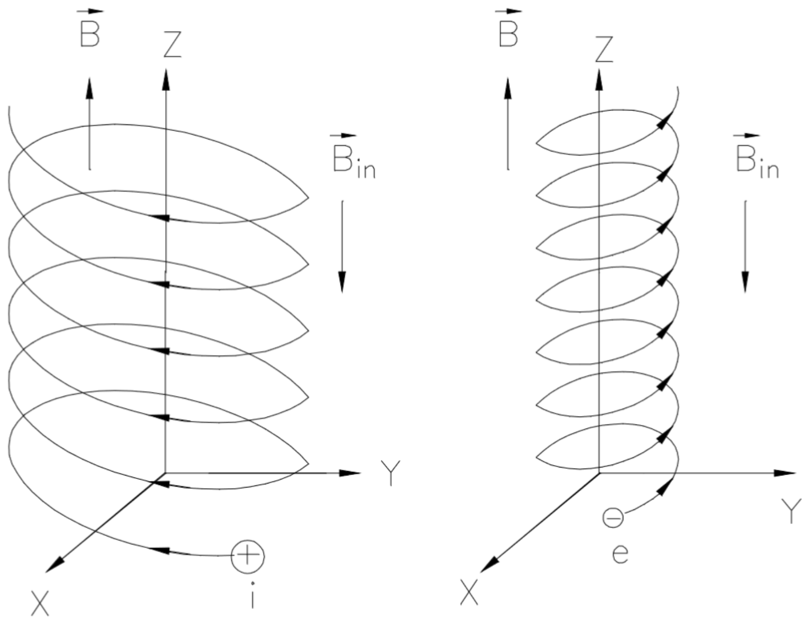
Outline

- Transport and anomalous transport
- Biasing experiment on STOR-M
- Conventional signal analyses (examples from STOR-M)
- Nonlinear signal analyses
  - Algorithm
  - Numerical Validation
  - experiments on STOR-M
- Experiments on TJ-II
- Experiments on EAST
- Other experiments?

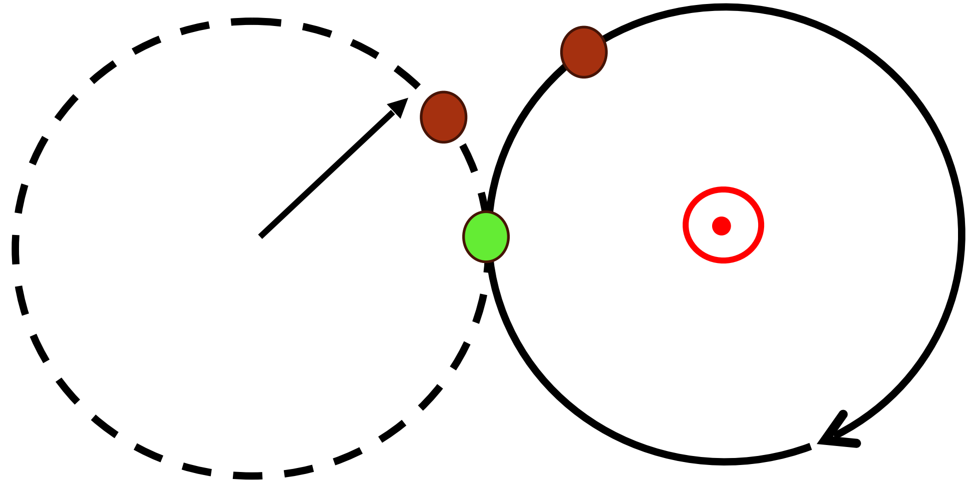
Transport

- Charged particle motion in a uniform magnetic field



- Confined by magnetic field lines with Larmor radius
- Diamagnetic
- Free to move along magnetic field lines

Collison causes cross-field transport



Before collision After collision

Radom walk step length:

Fick’s Law and diffusion coefficient

Fick’s law:

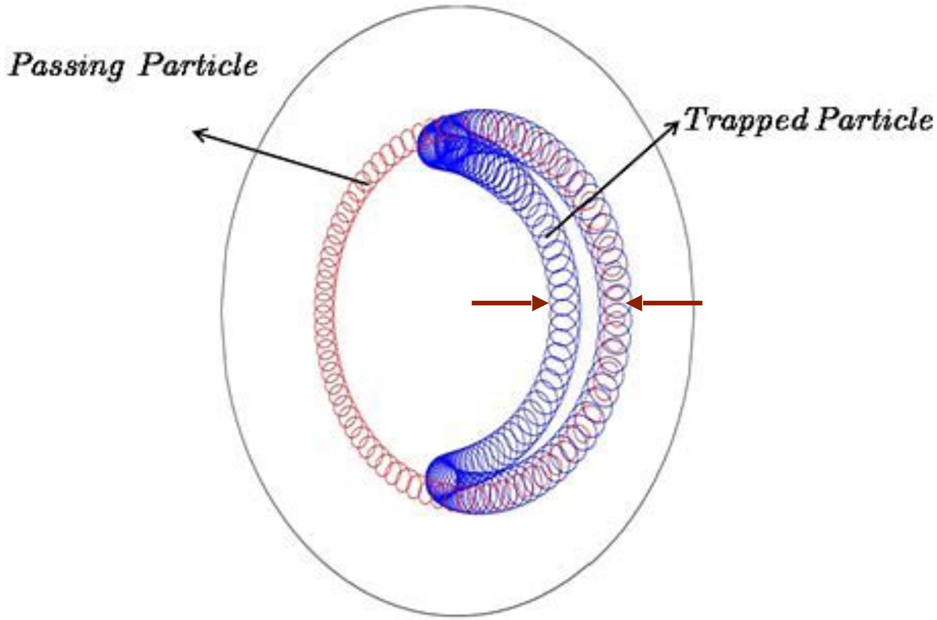
- Particles diffuse from high to low density region
- The diffusion coefficient is

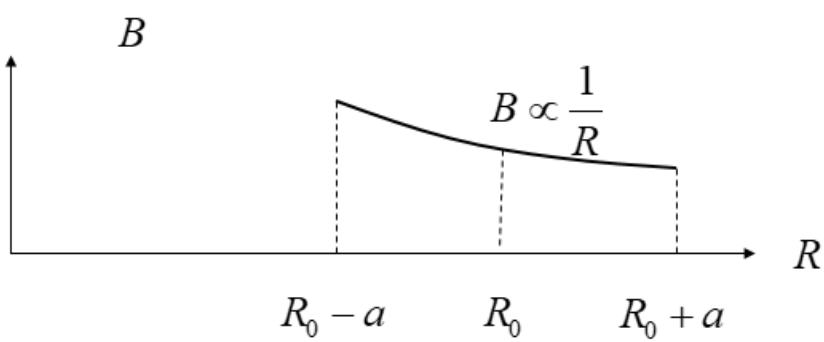
Classical transport

Coulomb collision theory gives

Neo-classical transport

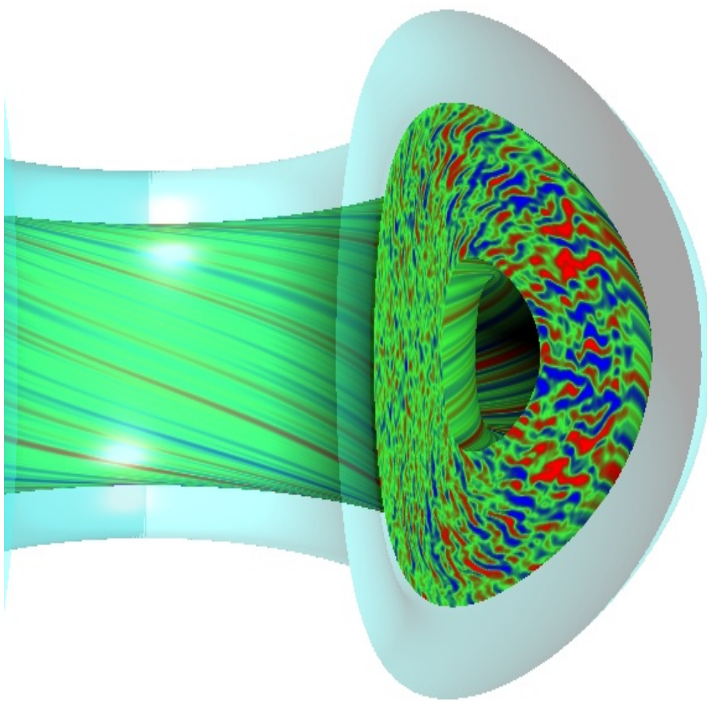
Banana orbit is formed due to mirror effects in tokamaks:





Anomalous transport

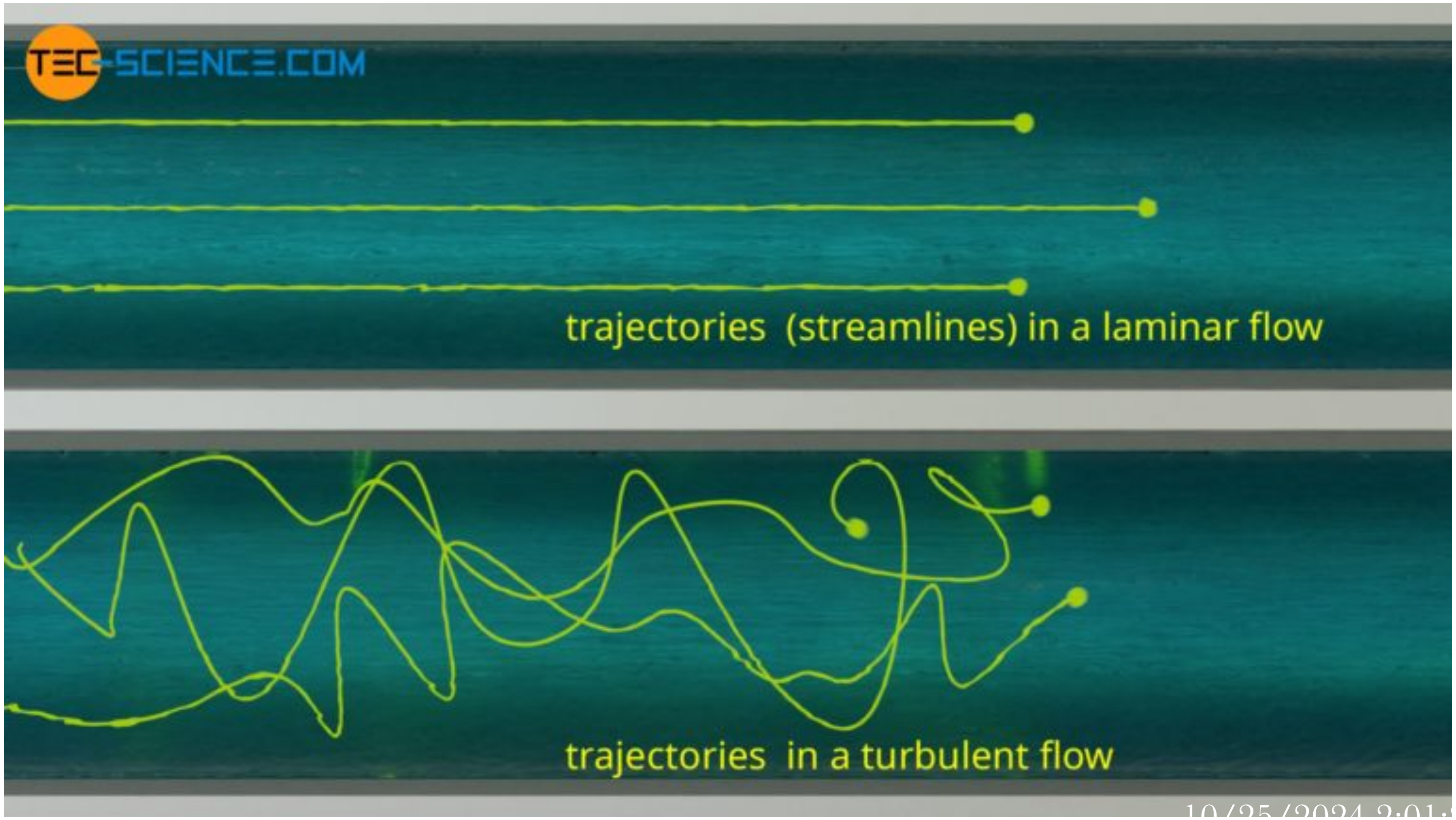
- Anomalous transport is caused by turbulence
- There is no theory for turbulence
- Modeling and simulation reveals turbulence



- Radom walk step length is now replaced by the turbulence eddy size
- Anomalous transport exceeding classical transport is hindering fusion energy development

Laminar flow and turbulent flow

- Laminar flow occur at low speed
- Turbulent flow occurs at high speed with chaotic streamlines



What causes turbulence?

- Reynolds number

Turbulence occurs when

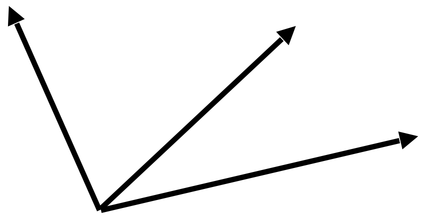
- High flow velocity
- Low viscosity

Drift type instabilities

- Cross field drift

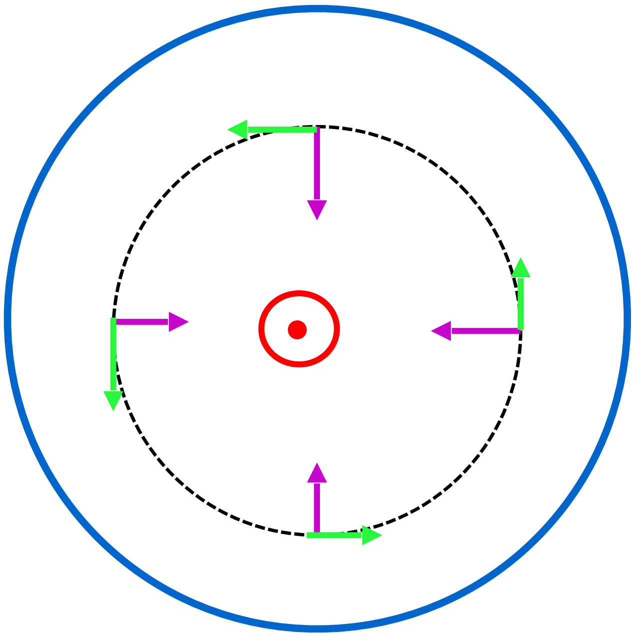
  In addition to magnetic force , another force is added

  This causes a drift perpendicular to both the force and the magnetic field



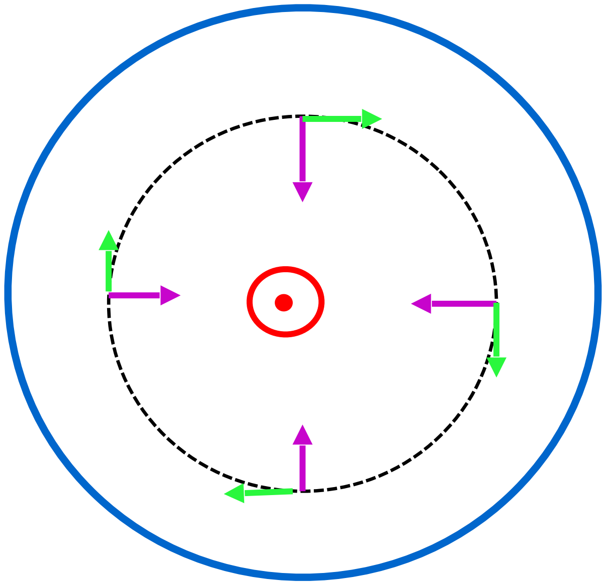
drift in tokamaks

- Radial electric force (assumed inwards)



Pressure Gradient Drift

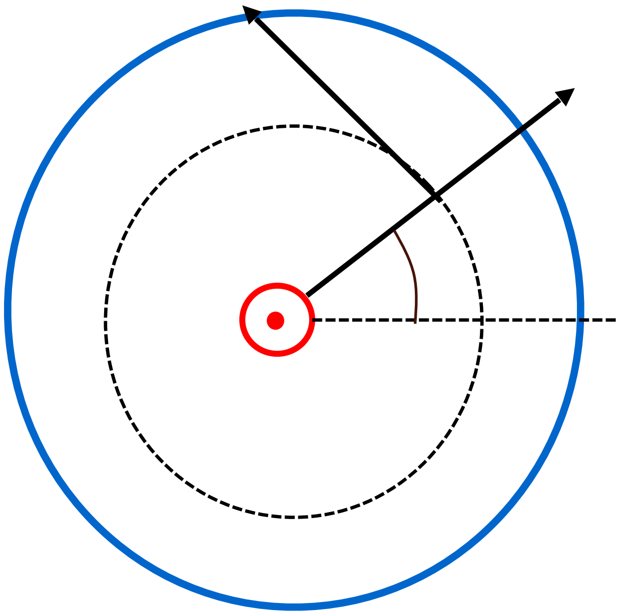
- Pressure force per particle



- Charge dependent drift direction
- Electron (ion) drift direction
- Diamagnetic drift

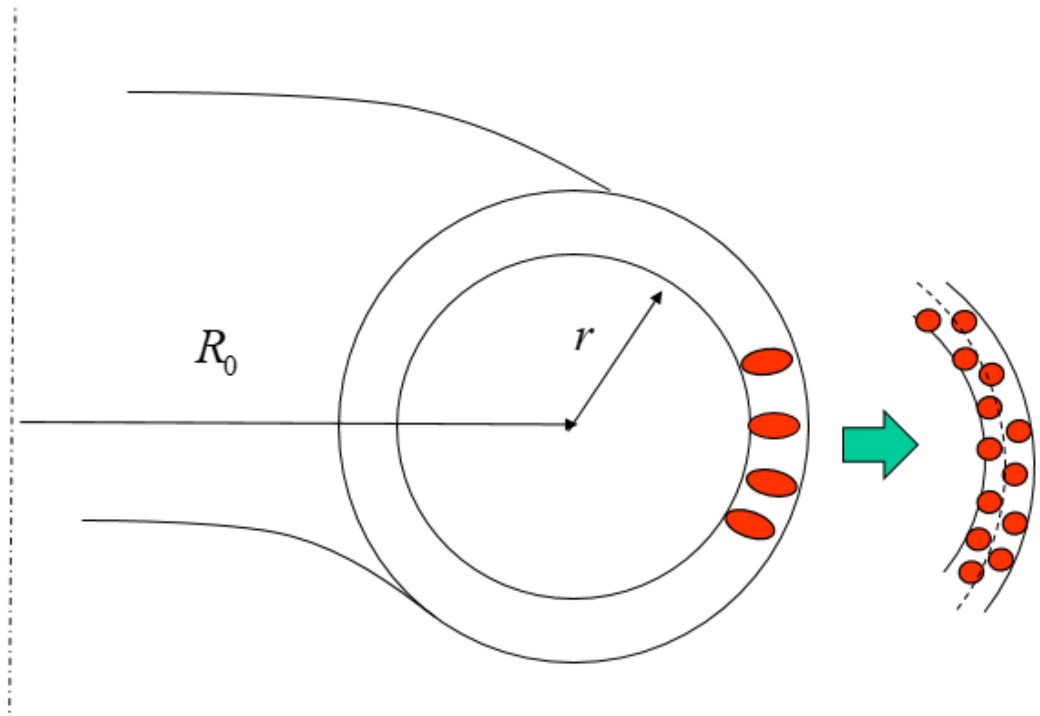
Radial Force Balance

- Force equation



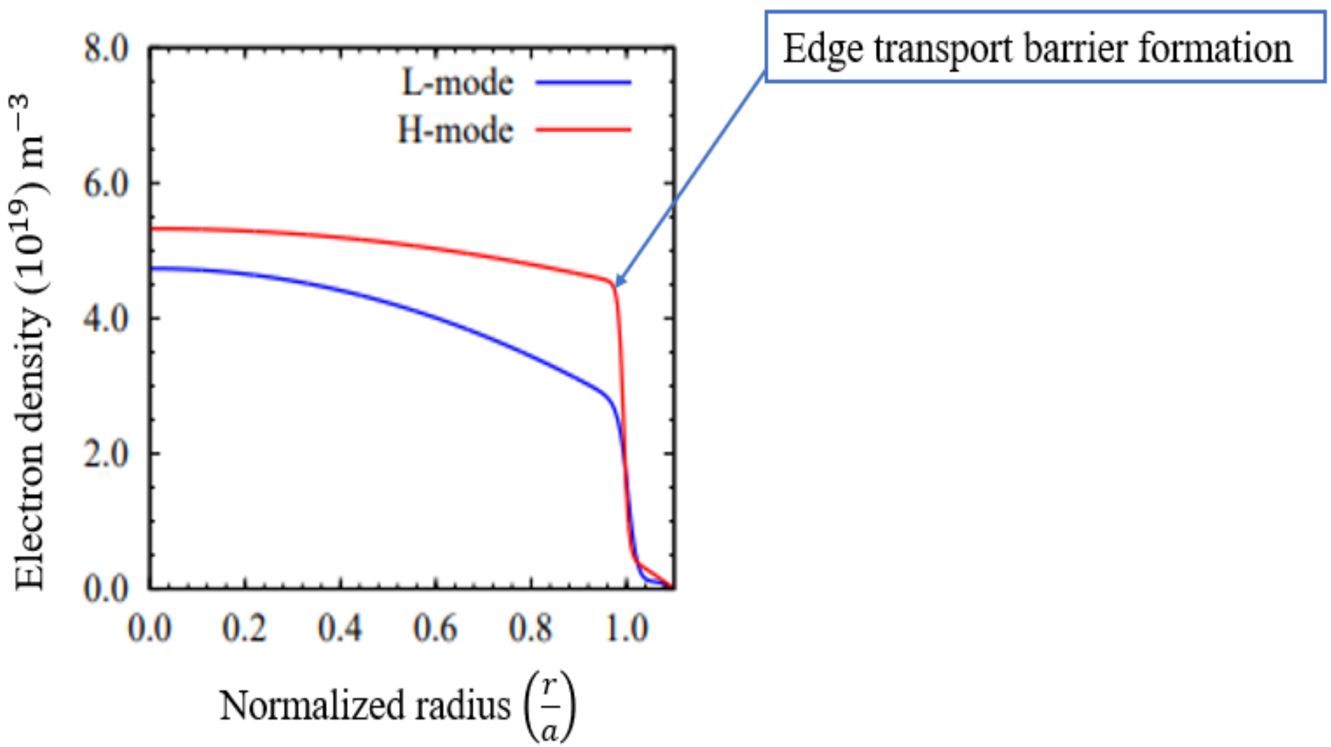
Turbulence Eddy Breakup

- Electrode biasing generate an flow shear
- Breaks up the turbulence eddies to smaller ones and improves the confinement



Pedestal Formation

- Helps to generate a poloidal flow shear
- Breaks up the turbulence eddies and improves the confinement



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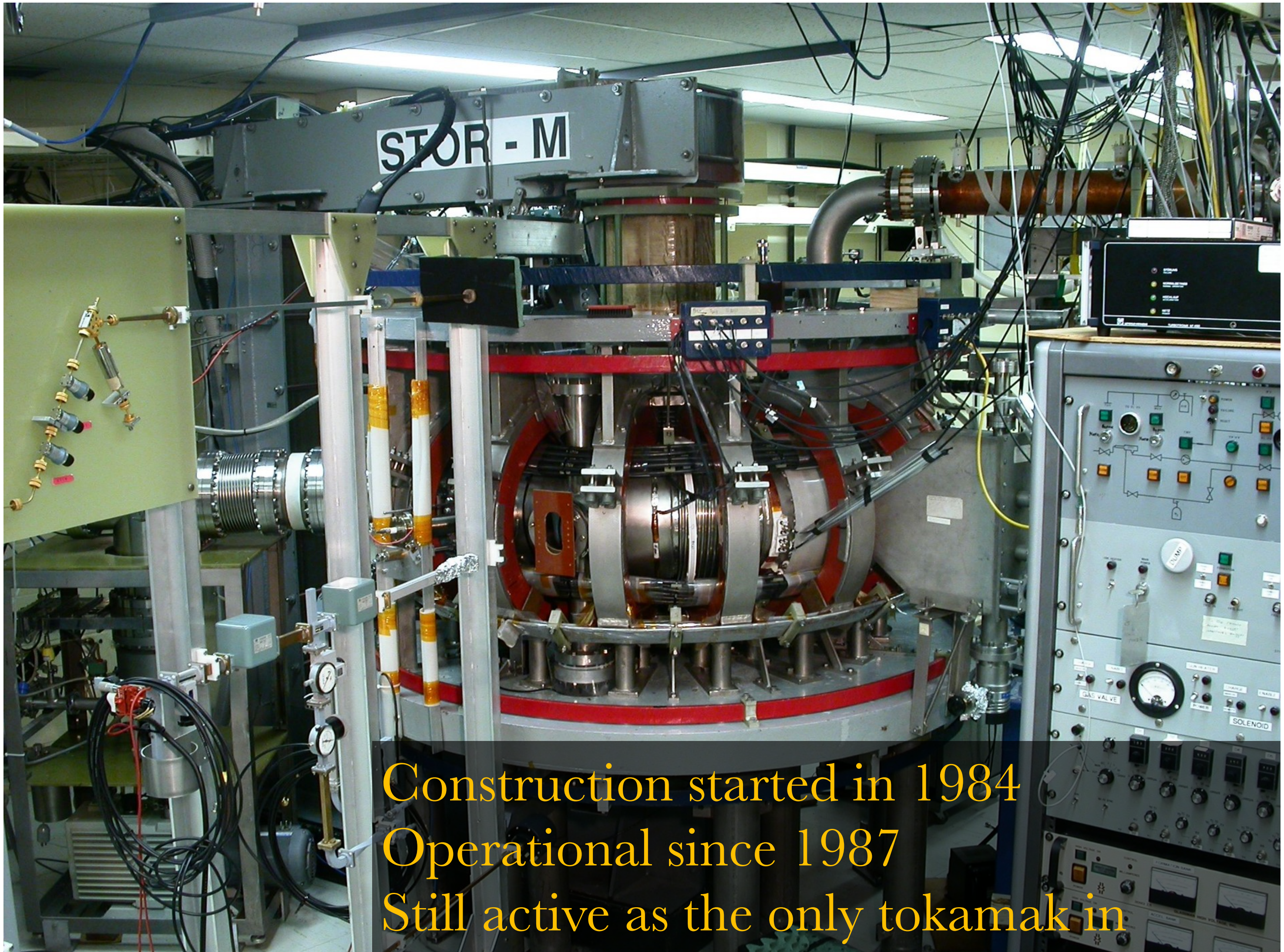
STOR-M



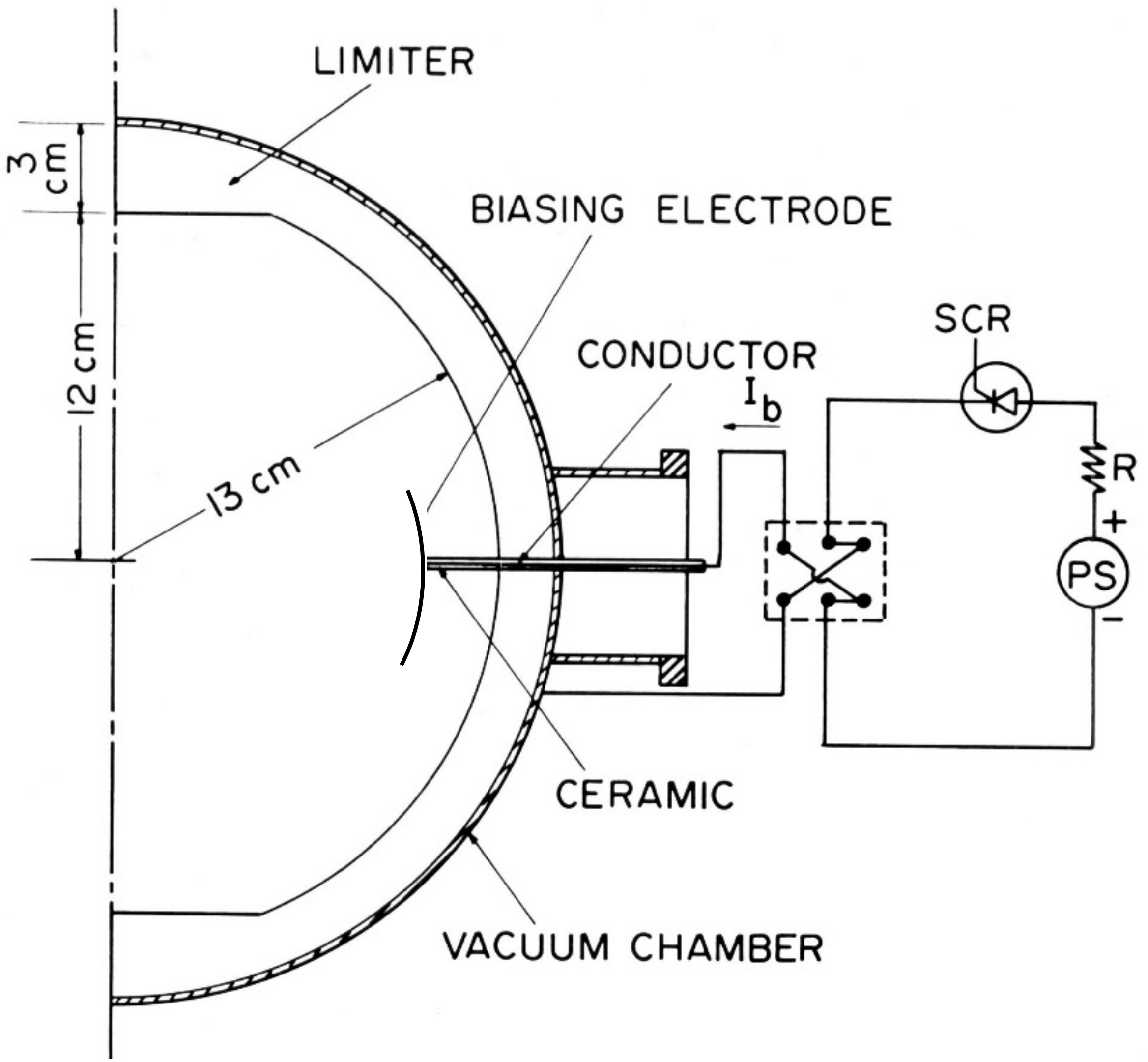
Construction started in 1984
Operational since 1987
Still active as the only tokamak in Canada

STOR-M Tokamak Parameters

Major radius	R	46 cm
Minor radius (limiter)	a	12 cm
Toroidal B field	B_ϕ	1 T
Plasma current	I_p	20-30 kA
Average electron density	n_e	$1 \sim 3\times\ 10^{13}\ \mathrm{cm}^{-3}$
Electron temperature	T_e	220 eV
Ion temperature	T_i	50~100 eV
Discharge duration	t_d	50 ms
Energy confinement time	τ_E	1~3 ms

Electrode Biasing Experiments

- DC Biasing



Transient Ohmic H-Mode

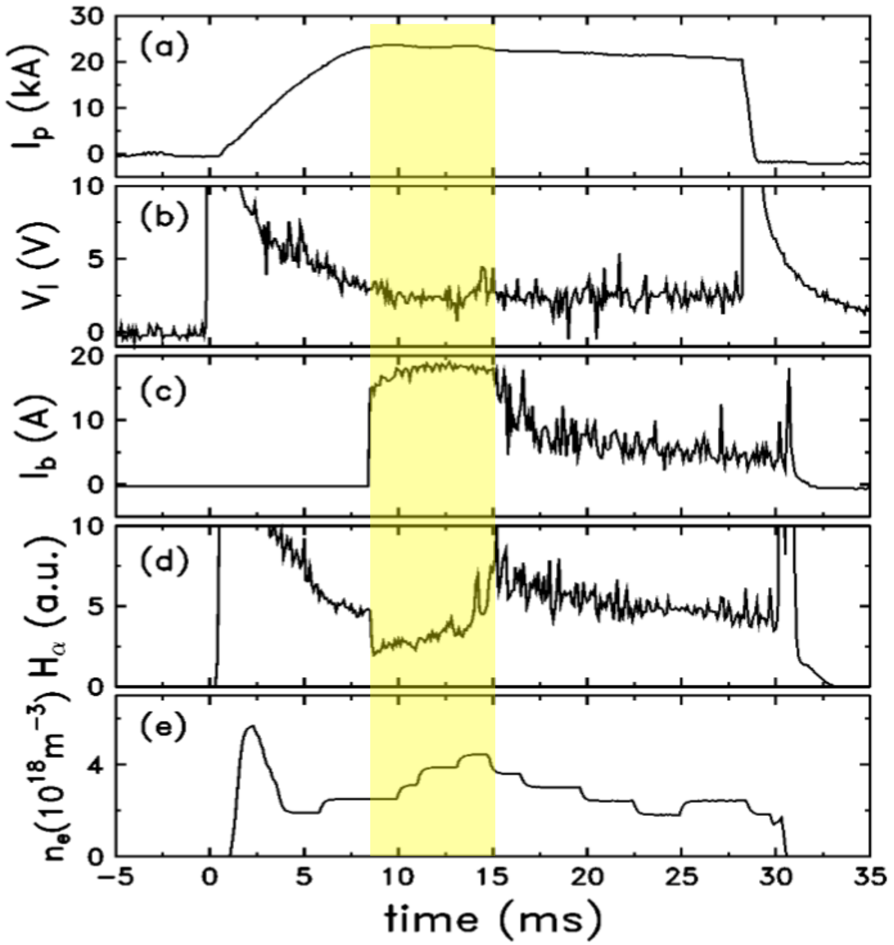


FIG. 3. Discharge history with biasing voltage −433 V; (a) plasma current; (b) loop voltage; (c) biasing current; (d) H_α radiation; (e) electron density. Shot No. 16 484.

Modification of floating potential

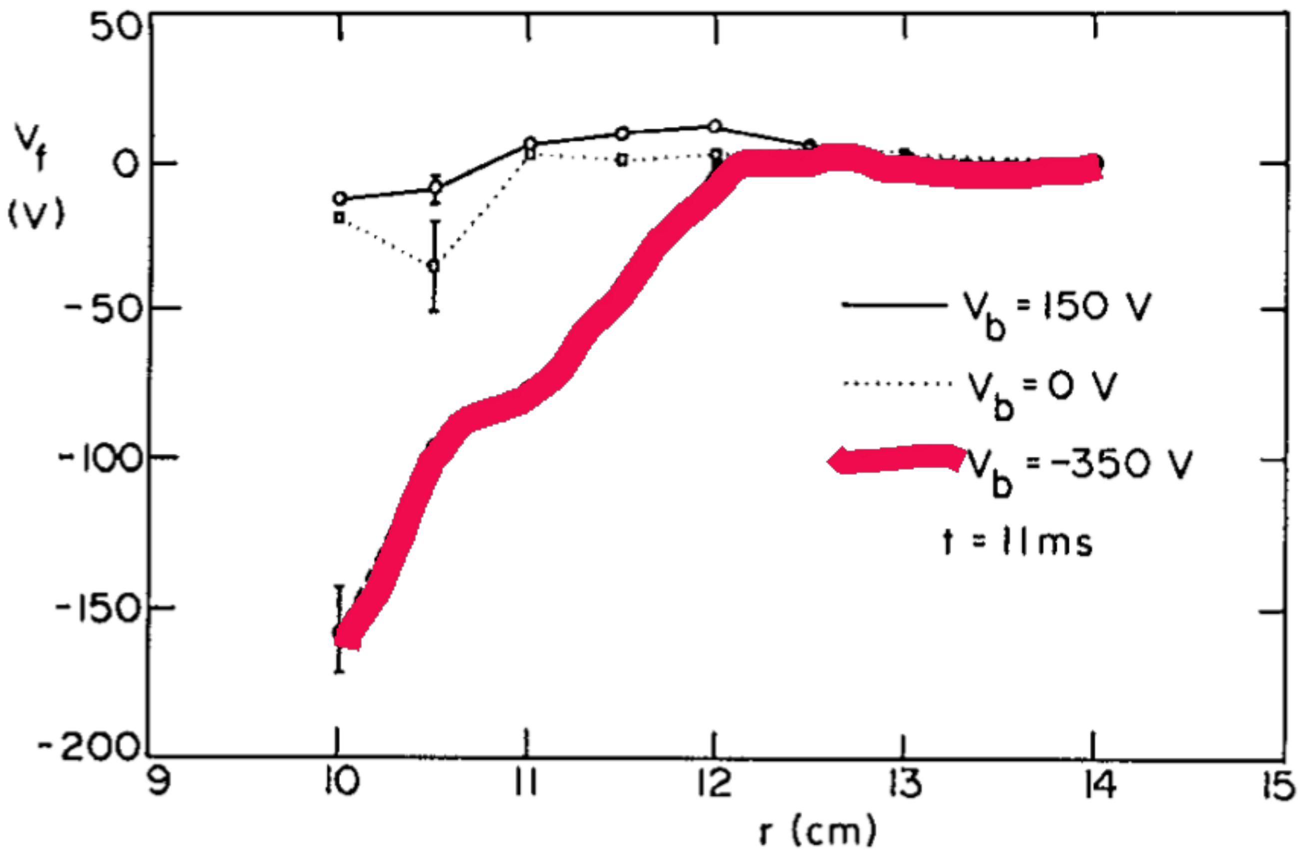


FIG. 4. Radial profile of floating potential at the equatorial midplane in an outboard region under three conditions: without bias, −350 V bias, and +150 V bias.

Modification of Density Profile

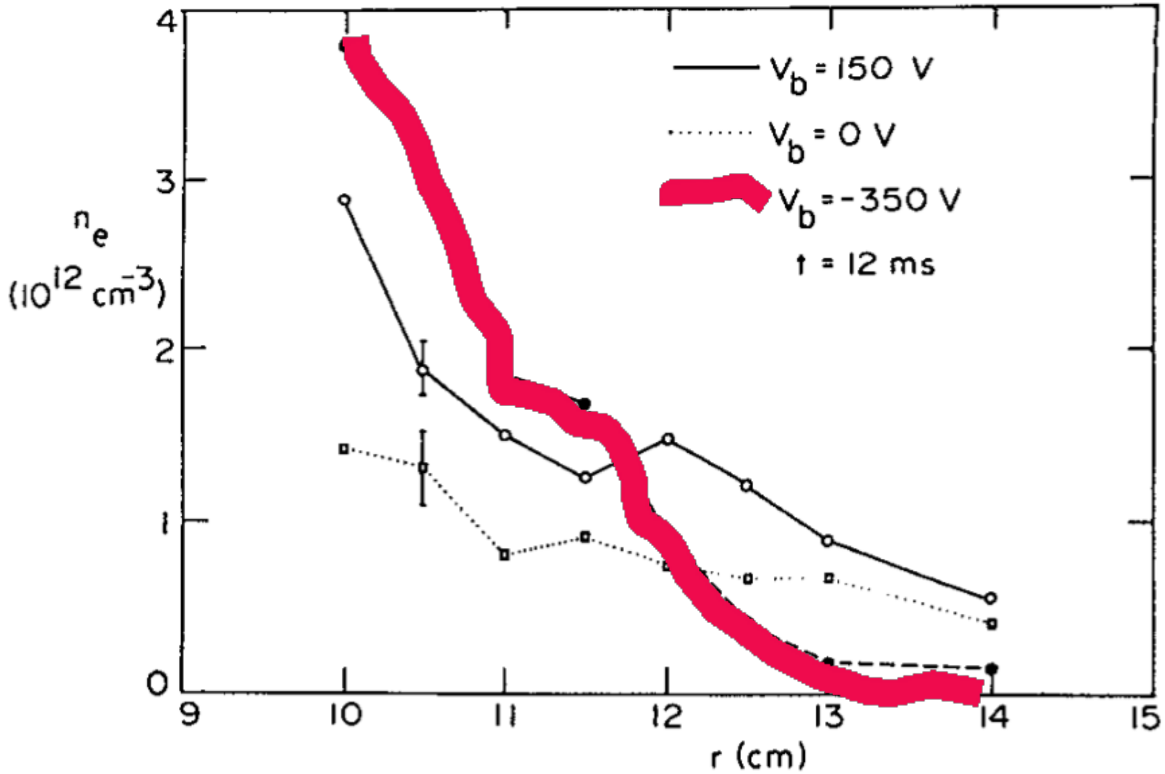
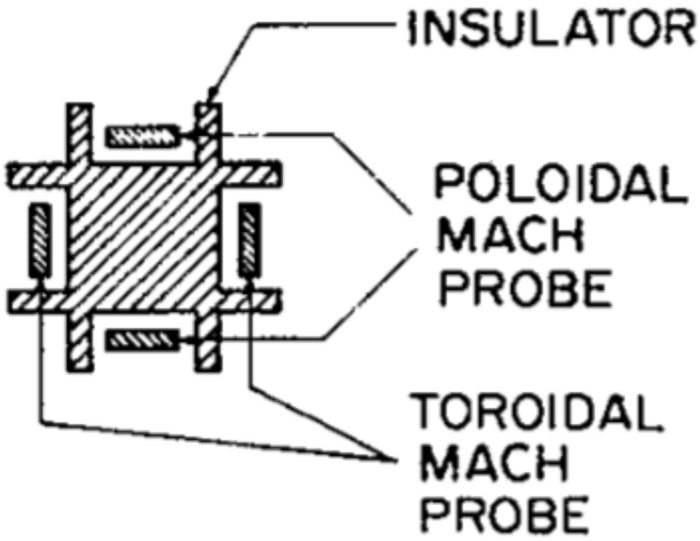


FIG. 5. Edge plasma density profile versus the minor radius under three conditions: without bias, −350 V bias, and +150 V bias.

Mach Probe

- Four-sided Mach probe



Toroidal Flow

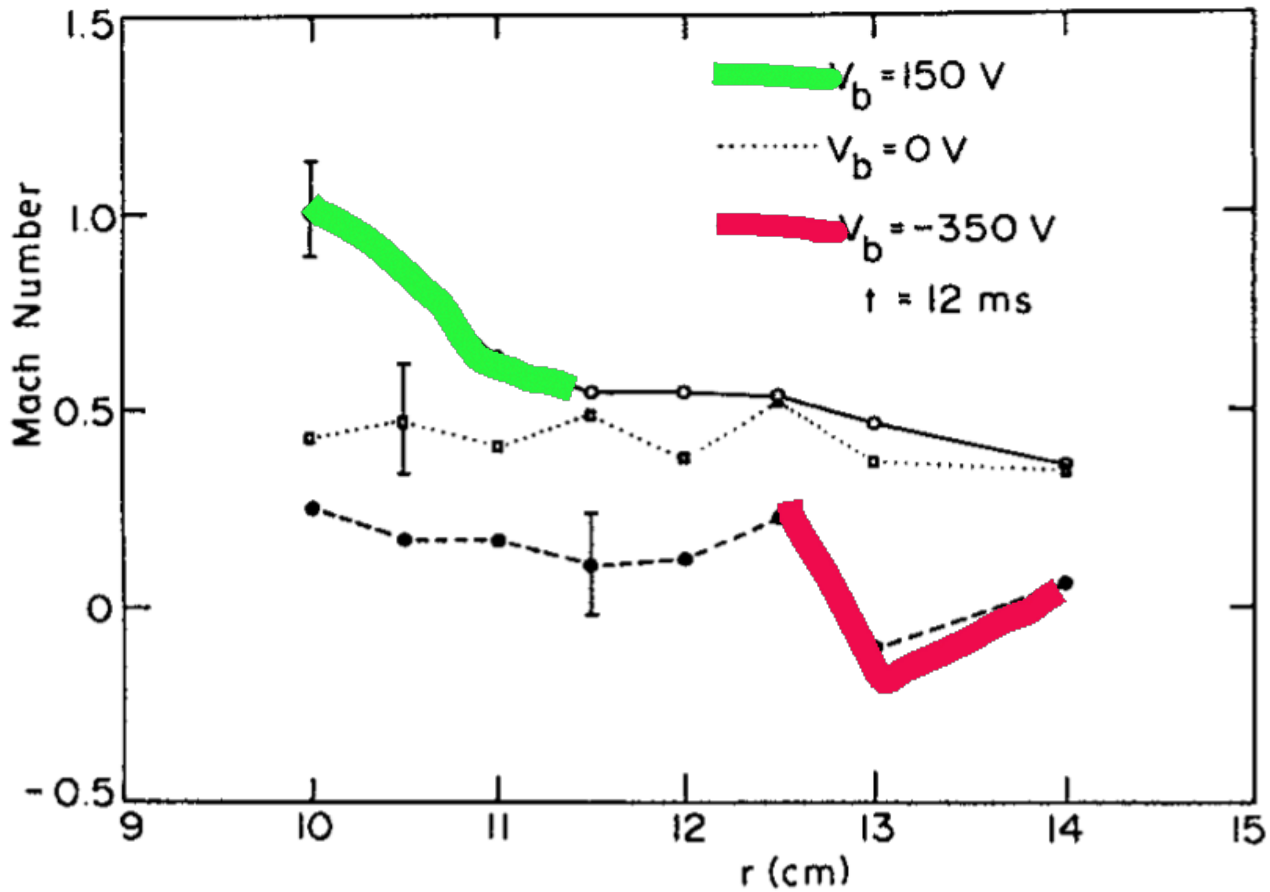


FIG. 8. Toroidal flow Mach number versus the minor radius for cases of positive (solid line), negative (dashed line), and no bias (dotted line).

Poloidal Flow

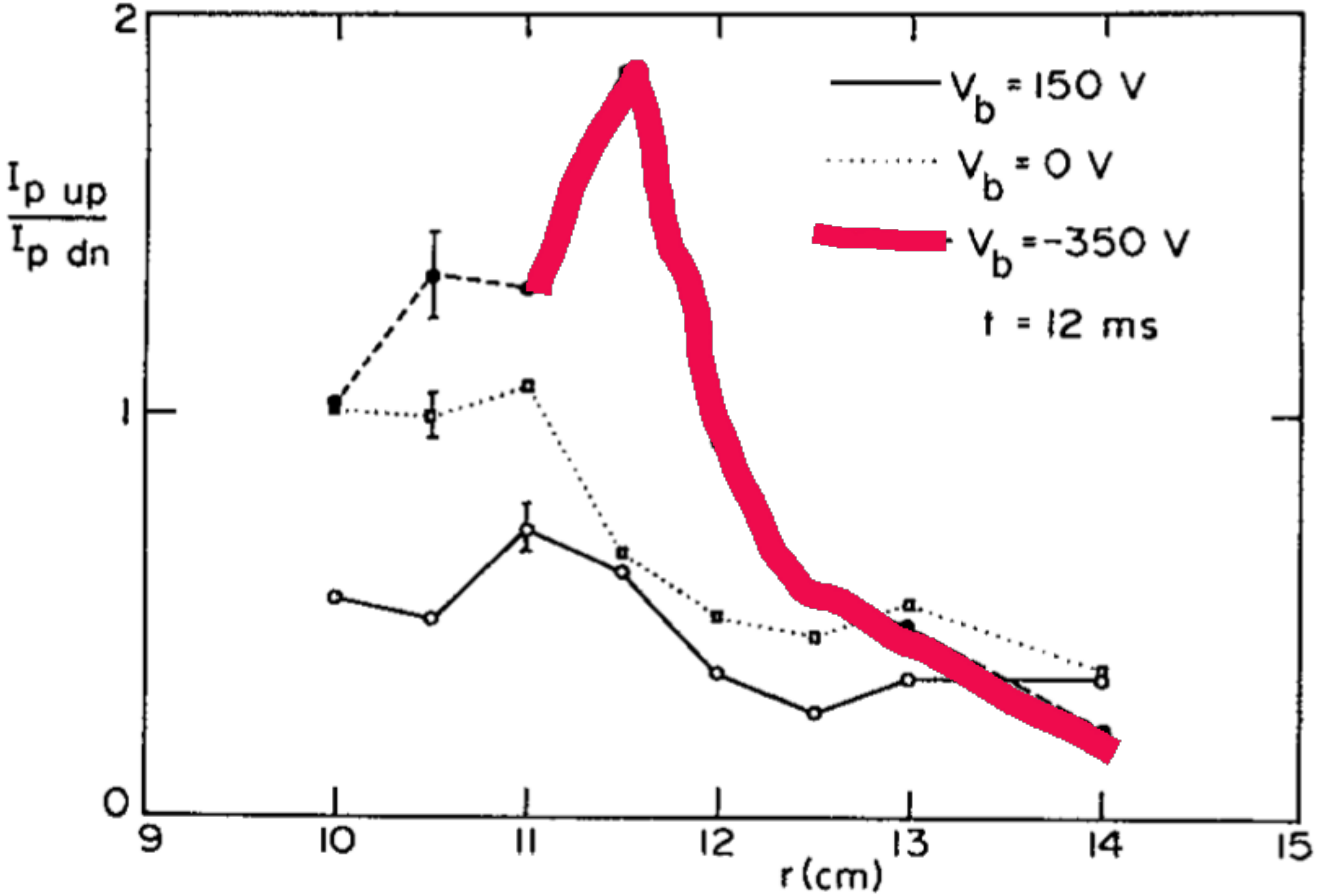
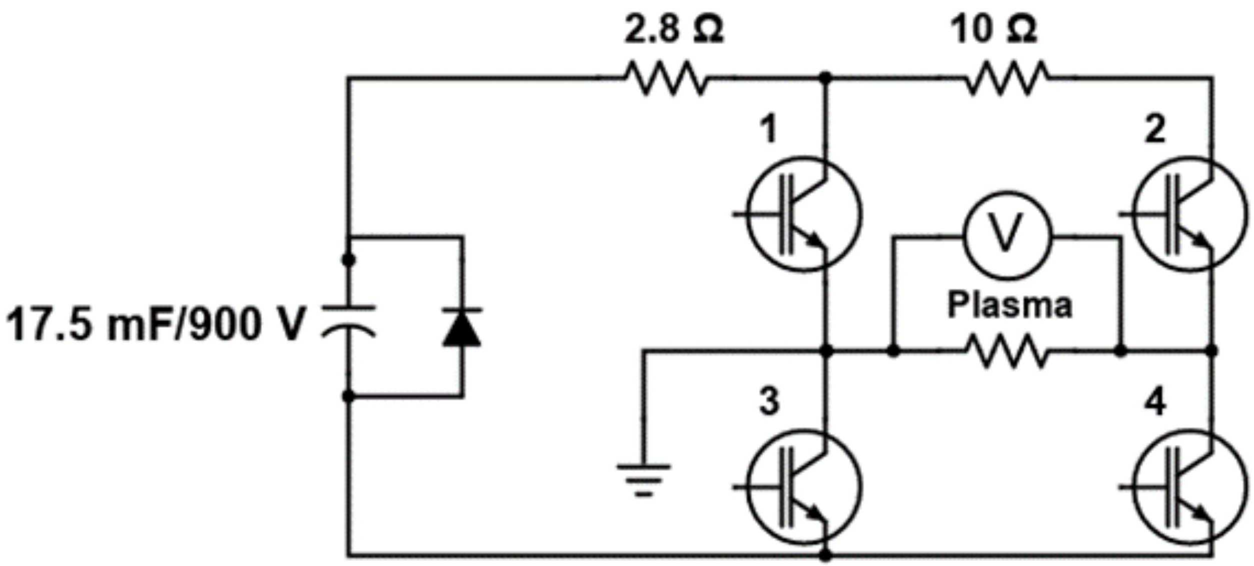


FIG. 9. Ratio of upstream to downstream poloidal ion saturation current versus the minor radius for cases of positive (solid line), negative (dashed line), and no bias (dotted line).

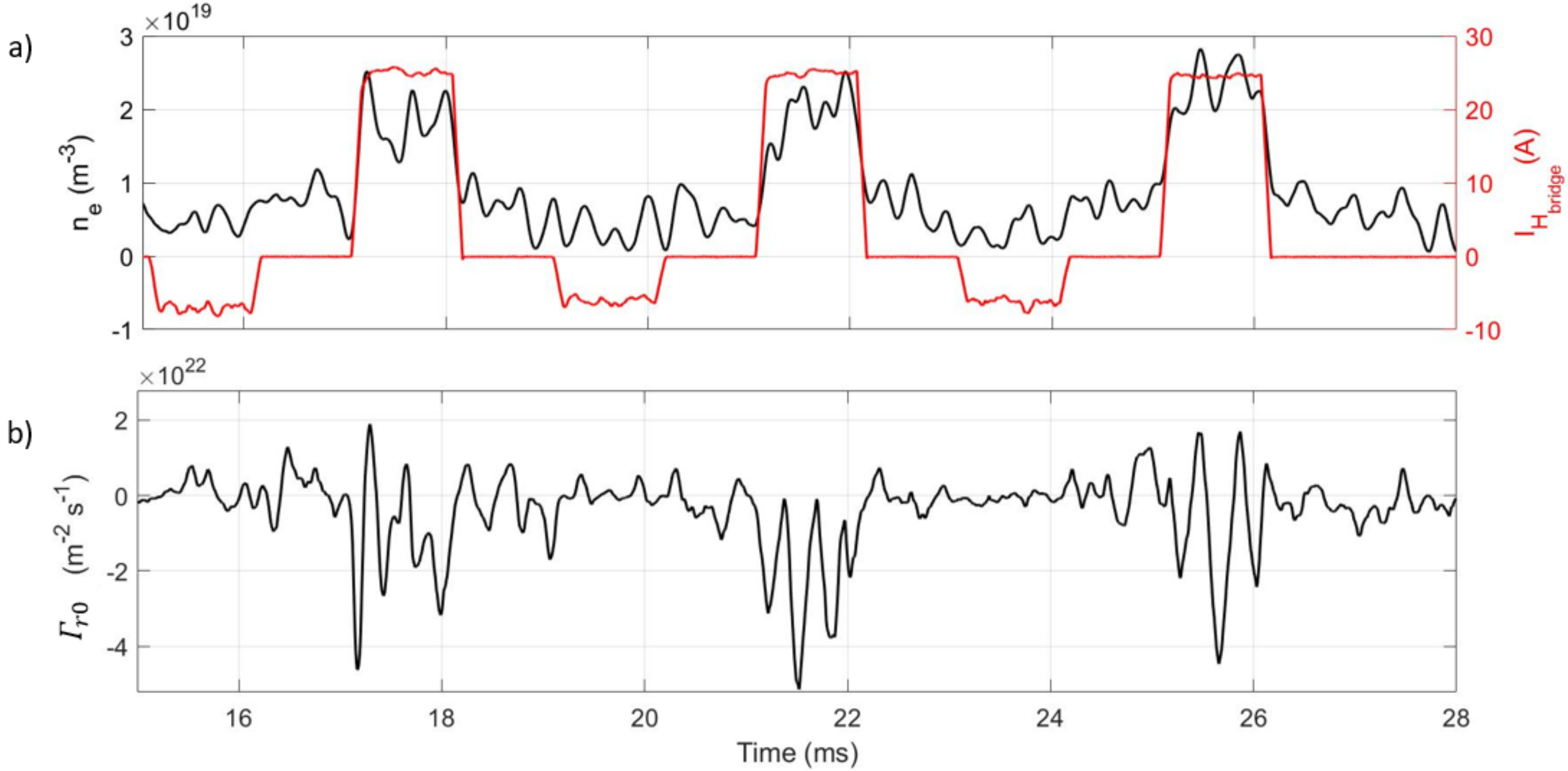
Other Biasing Waveforms

- FPGA controlled H-bridge
- Programable bipolar pulses



Bipolar Biasing

- Modified density
- Modified particle flux



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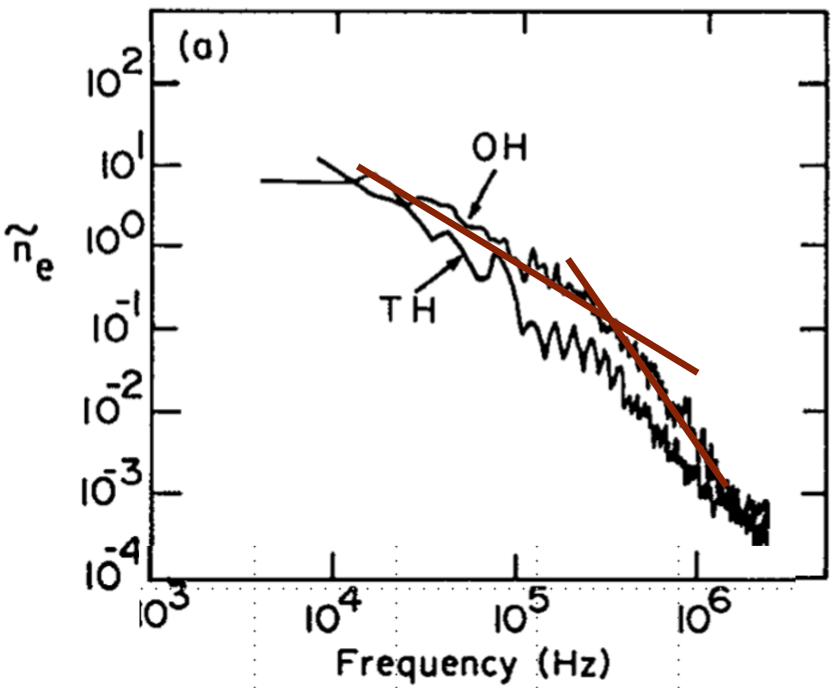
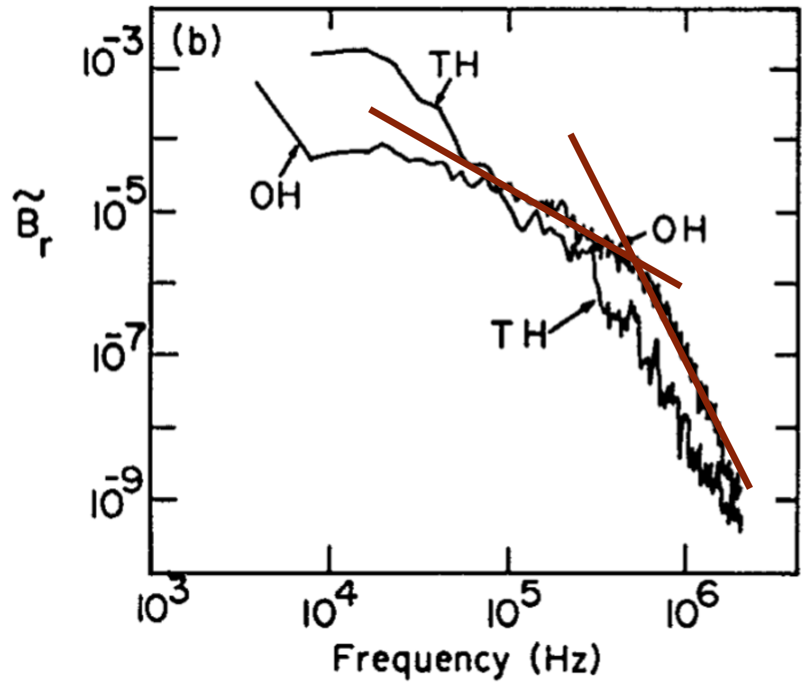
Conventional Signal Analyses

- Auto- and cross-correlation (time domain)
  - Time scale and spatial spread of turbulence eddies
  - Phase propagation direction and velocity
- FFT analyses (frequency domain) or wavelet
  - Power spectra (magnitude only)
  - Phase propagation direction
  - Coherence (strength only)
- Bicoherence
  - Mode coupling strength (no directions)

Power Spectra of Fluctuations

- FFT (DFT)

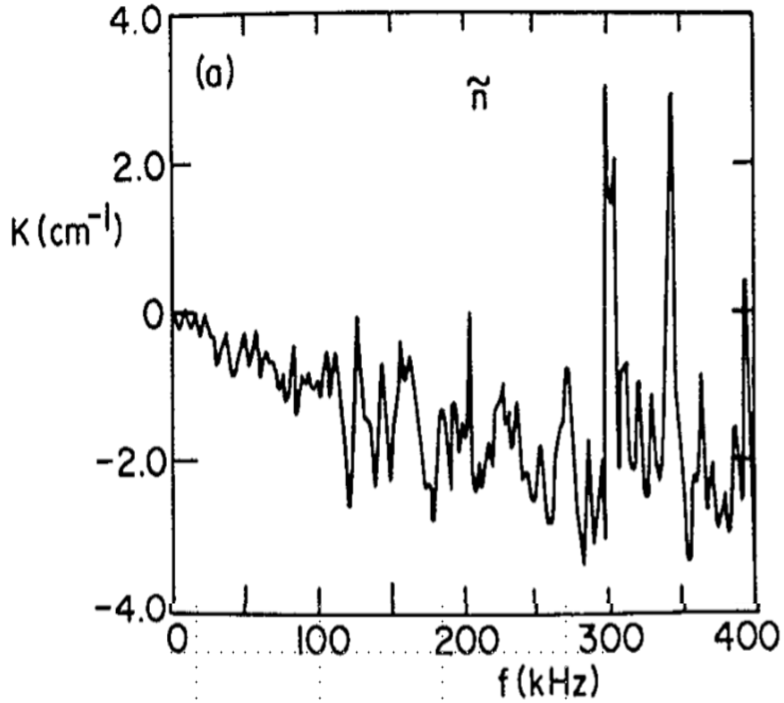


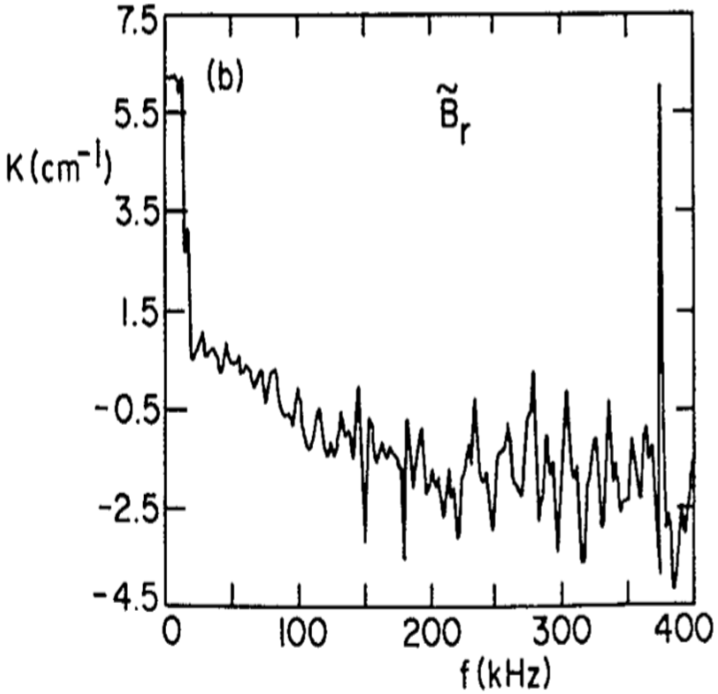


  - Broad band
  - Different slopes in low and high frequency ranges

Dispersion Relation

- Phase difference of two probes separated in space (poloidal for the case below)





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Nonlinear Cross-Correlation

- Plasma is governed by nonlinear equations
- Solutions are not unique
- Solutions are sensitive to plasma parameters, initial/ boundary conditions, perturbations
- Jumping between solutions (H-L-H transitions)
- Chaotic or self-organized system
- A different method may be used to reveal turbulence energy propagation direction

Conditional nonlinear cross-correlation

- Complex nonlinear (chaotic) systems (in multi-dimensional phase space)
  - Self-Organized Critical (SOC) systems
  - Bifurcation of chaotic systems and formation of attractors (L-H transition)
- Examples:
  - Transfer Entropy (B.Ph. van Milligen, *et al.* NF **54** (2014) 023011) - entropy (energy) transport direction
  - Our algorithm (W. X. Ding, *et al.* PRL **79** (1997) 2458)

Algorithm

- Time series of two signals x_k and $y_{k'}$, (e.g., floating potentials of two radially separated Langmuir probe pins)
- Form a series of *M*-dimensional vectors using delay signals (*M*: embedded dimension to be determined)

  - *k* is related to the auto cross-correlation time (number of data points to be skipped)
  - *n* is another constant integer (related to the number of data points to be skipped to form next vector)
  - N number of vectors in the ensemble for statistics

	3	4	5	6	7	8	9	10	11	12	13	14									

Algorithm

- Define Euclidean "distance" between two "points" :

$$\epsilon_{xij}^{M} = \left|\mathbf{x}_i^M - \mathbf{x}_j^M\right|, \qquad \epsilon_{yij}^{M} = \left|\mathbf{y}_i^M - \mathbf{y}_j^M\right|$$

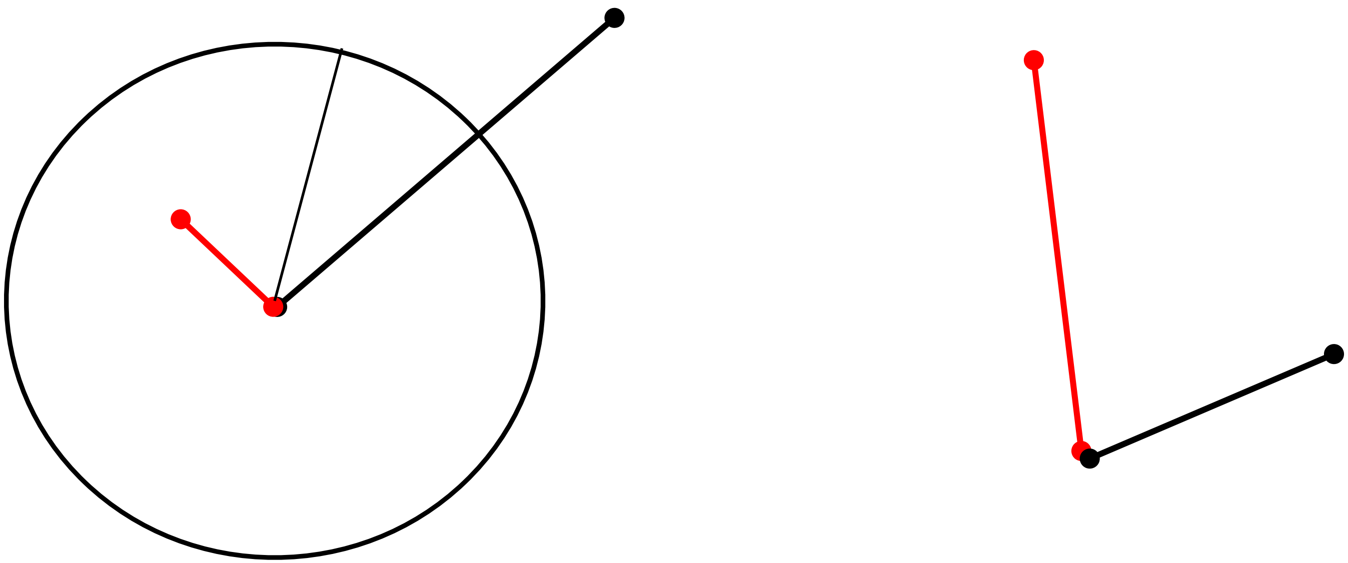
- Conditional probability $P\left(\mathbf{y}_i^M \mid \mathbf{x}_i^M\right)$ (N. H. Packard, et al., PRL 45 (1980)712)
  - probability finding $\mathbf{y}_i^M$ for fixed $\mathbf{x}_i^M$
  - If they belong to independent subsystems, $P\left(\mathbf{y}_i^M \mid \mathbf{x}_i^M\right)$ should depends only on $\mathbf{y}_i^M$ itself ($P\left(\mathbf{y}_i^M\right)$)

Algorithm

- Mean conditional dispersion

$$\sigma_{xy}^{M}(\epsilon) = \sqrt{\frac{\sum_{i\neq j} \left|\mathbf{y}_i^M - \mathbf{y}_j^M\right|^2 H\left(\epsilon - \left|\mathbf{x}_i^M - \mathbf{x}_j^M\right|\right)}{\sum_{i\neq j} H\left(\epsilon - \left|\mathbf{x}_i^M - \mathbf{x}_j^M\right|\right)}},$$

where *H* is the Heaviside function

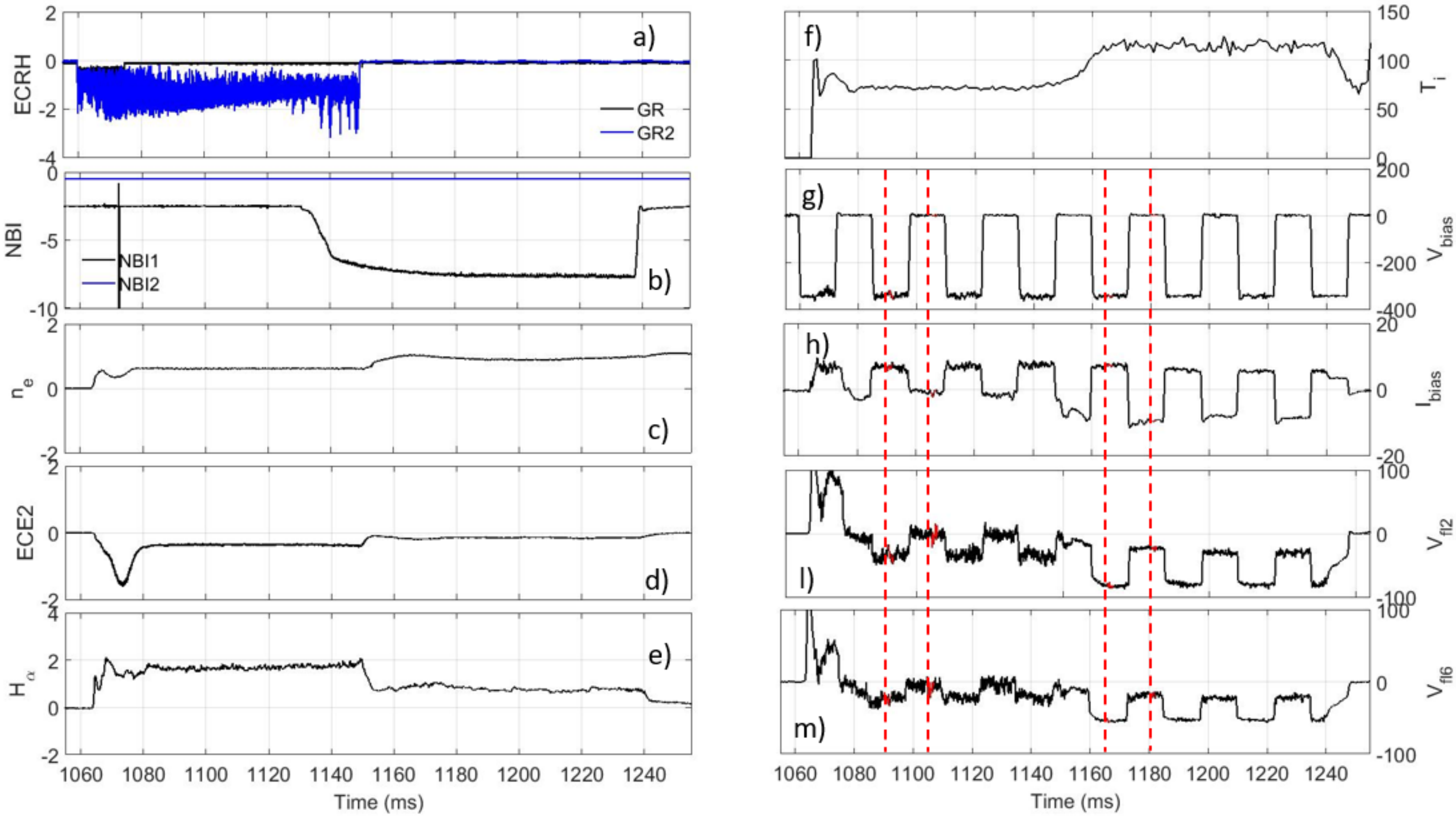


Algorithm

- For fixed dimension *M*, $\sigma_{xy}^{M}(\epsilon)$ is a function of ϵ, a normalized curve can be drawn
- For a series of *M's,* many dispersion curves can be drawn
- When *M* is large enough, they merge to almost the same dispersion curve.
- The merged curve can be used to derive a directional correlation coefficient (see the following example)

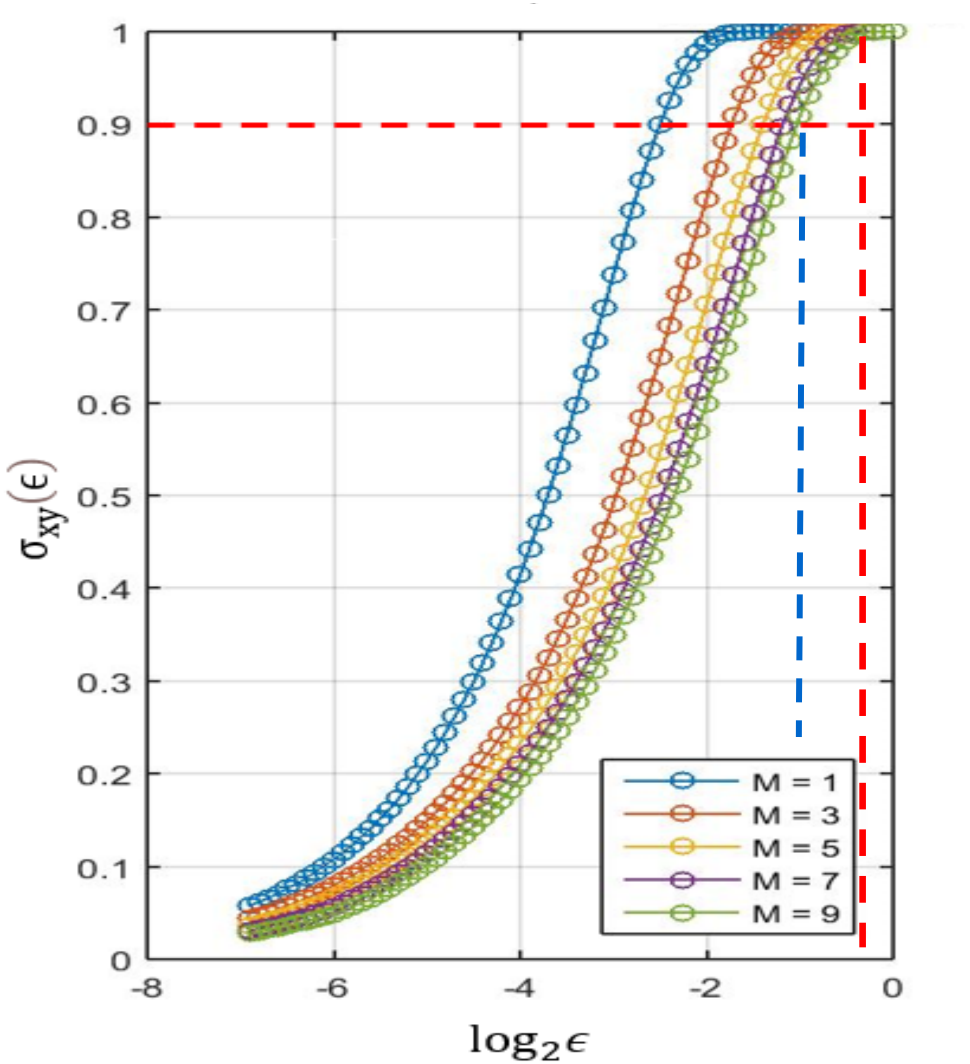
Sample data (dispersion curves)

- TJ-II discharge #43823 with electrode biasing pulses (-350 V)



Sample data

- Dispersion relation and correlation coefficient



$$\boxed{g_{xy} = \lim_{M\to\infty}\left(\frac{\epsilon_{90}}{\epsilon_{max}}\right)}$$

- Similarly, find also g_{yx}
- The asymmetry determines the energy propagation direction
  - $g_{xy} > g_{yx}$: propagate from x to y
  - $g_{yx} > g_{xy}$: propagate from y to x

Algorithm

- Correlation asymmetry

- Nonlinear cross-correlation direction

Numerical Validation

- Van der Pol coupled oscillators

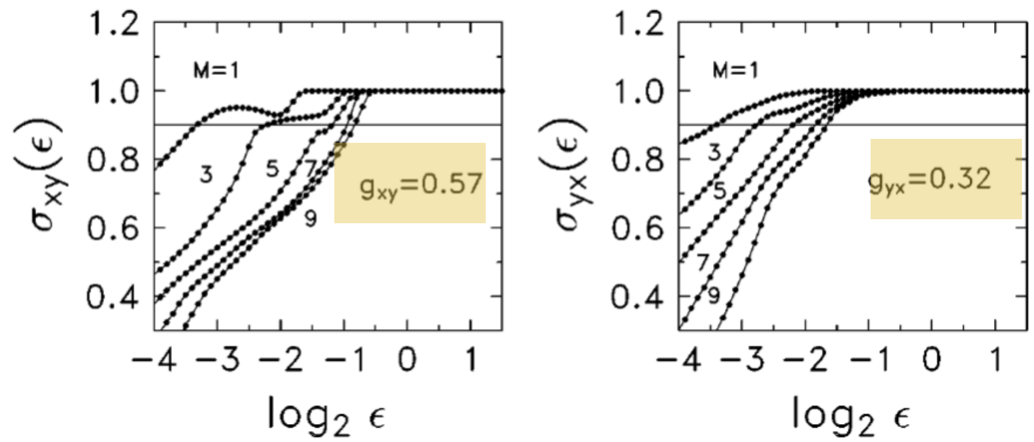


FIG. 1. Conditional dispersion curves are calculated based on simulated data from Eqs. (2) and (3) for $\alpha = 2.5, \beta = -0.5, N = 1000$ without noise.

Electrode biasing triggered ohmic H-mode in STOR-M

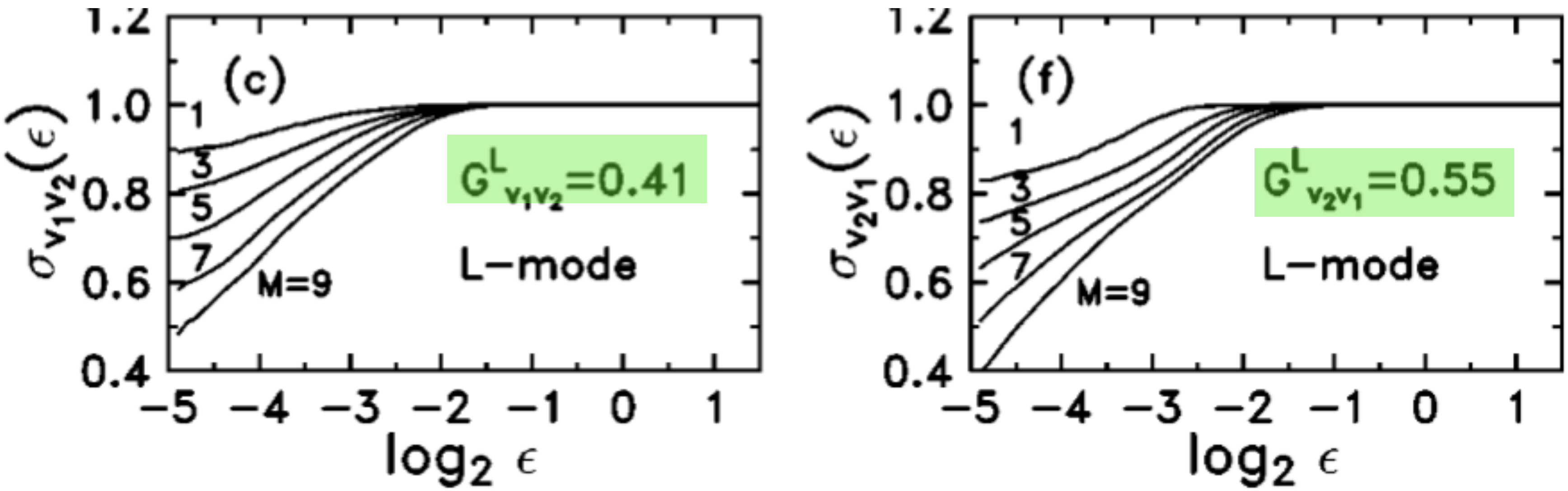


FIG. 4. *L*-mode experimental results. v_1 is the plasma potential at $r_1 = 0.83a$, v_2 at $r_2 = 0.95a$. Shot No. 16 542.

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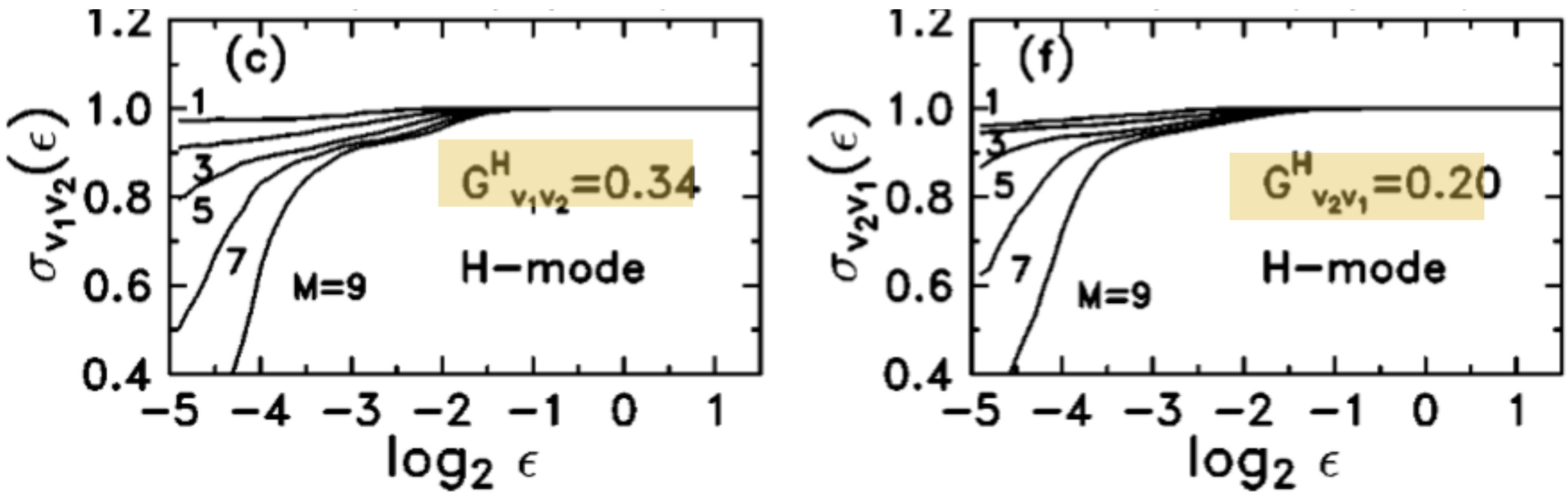

FIG. 5. *H*-mode experimental results. v_1 is the plasma potential at $r_1 = 0.83a$, v_2 at $r_2 = 0.95a$. Shot No. 16 484.

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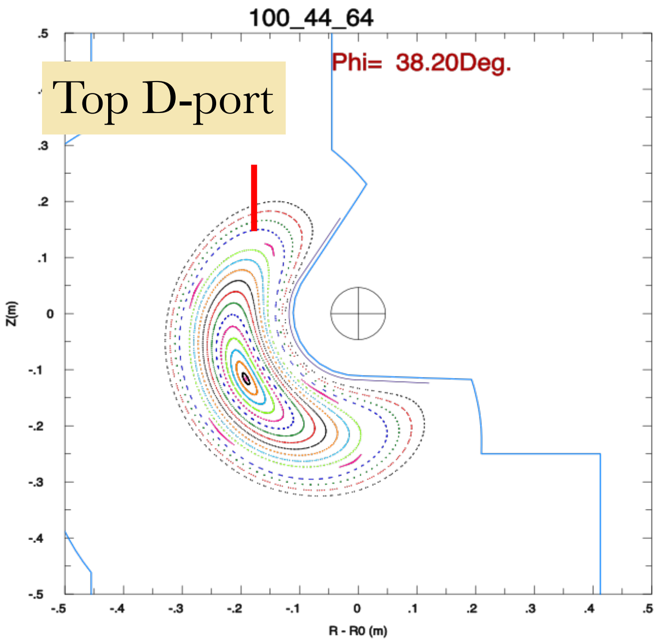
TJ-II experiments

- TJ-II (stellarators)
  - Medium size, well diagnosed device, good machine time accessibility
  - Lots of existing data already analyzed by different techniques



Experiments on TJ-II

- Continuous bipolar pulsed electrode biasing
  (-350 V, 0, +150V, 10 ms each voltage)
- Constant DC voltages
- Stationary rake probe (top D port) with flux measurement on the tip
- ECRH and NBI heated plasma
- Data obtained

Experiments on TJ-II

- Other spatially resolved fluctuation data can be downloaded as well
  - ECE (temperature)
  - HIBP (Potential, temperature)
  - Magnetic probes

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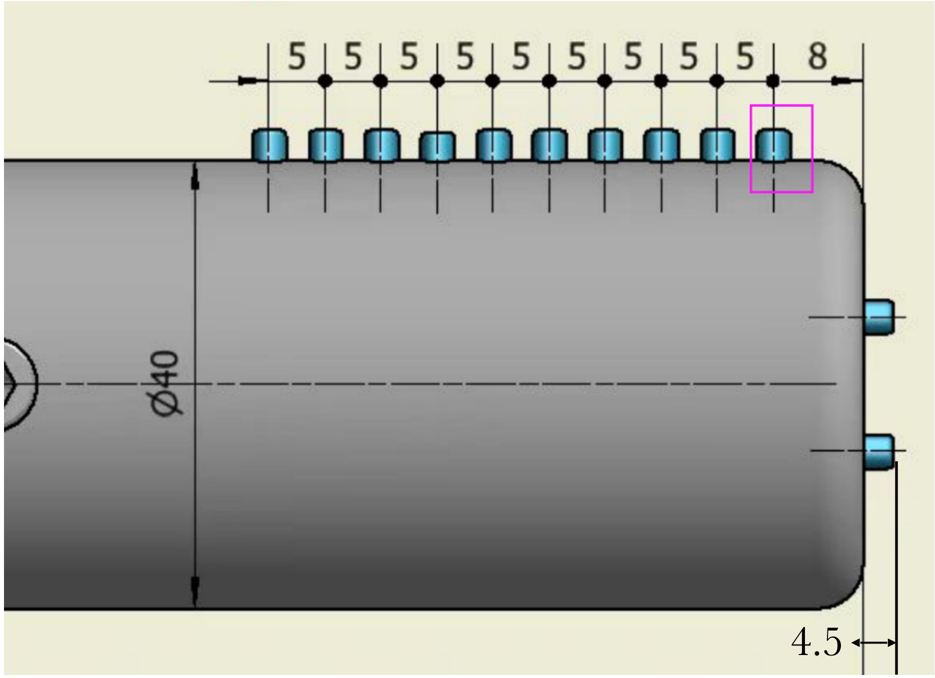
EAST Experiments

- Installed, some data obtained



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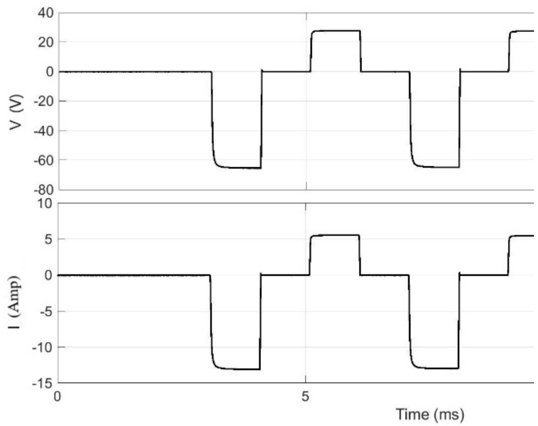
Future Experiments

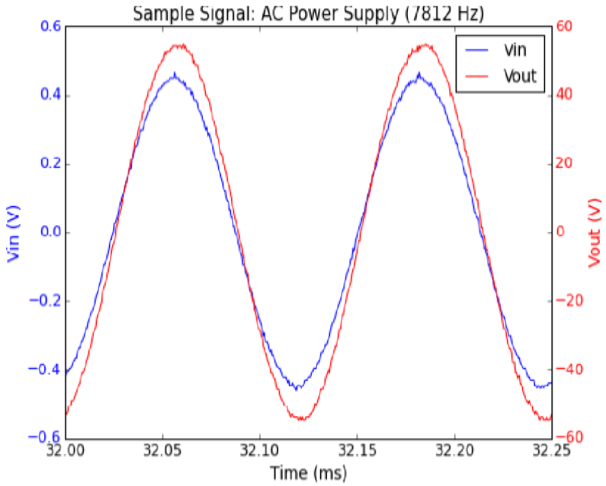
- STOR-M

  Small size, flexible device, full control of machine time

  Many biasing schemes (DC, AC, bipolar pulsed)

  Other H-mode triggering (CT, TH,RMP,...)





1-50 kHz, 120 V

- J-TEXT, KTX?

Summary

- Transport basics and drift type turbulence presented
- STOR-M biasing experiment reviewed
- Nonlinear cross-correlation techniques/ algorithm introduced with numerical validation and experimental results
- Experiments on EAST is ongoing
- Further experiments on other machines and for other signals can be explored

Thank you
for your attention!